2D Index Laws and the Zero Exponent

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Simplify each product using the First Index Law, leaving your answer in index form:

(a) $3^5 \times 3^4$ (b) $7 \times 7^6 \times 7^2$ (c) $x^4 \times x^7$ (d) $4a^3 \times 5a^5$

Q2. Simplify each quotient using the Second Index Law:

(a) $5^{11} \div 5^7$ (b) $\dfrac{2^{15}}{2^9}$ (c) $\dfrac{m^{10}}{m^4}$ (d) $\dfrac{24x^8}{6x^3}$

Q3. Simplify using the Power of a Power Law and the Zero Index Law:

(a) $(2^4)^3$ (b) $(m^5)^4 \div m^{12}$ (c) $9x^0 - (9x)^0$ (d) $5^0 + (-7)^0 - 2(4^0)$

Q4. Simplify completely and evaluate each numerical result:

(a) $\dfrac{3^4 \times 3^5}{3^6}$ (b) $\dfrac{(2^3)^4}{2^7}$ (c) $\dfrac{5^3 \times (5^2)^3}{5^7}$ (d) $\dfrac{4^5 \times 2^3}{2^{10}}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A cloud storage server in Sydney holds 2^{14} gigabytes of video files. If each video file is 2^3 gigabytes in size, express the total number of video files as a power of 2 and as a whole number.

Q6. A yeast culture in a food science laboratory triples in cell count every hour. Starting from 3^2 cells, how many cells are present after 5 hours? Write your answer in index form and as a whole number.

Q7. A cube has side length 2^3 cm. Express (a) the area of one square face and (b) the volume of the cube as powers of 2, then evaluate both.

Q8. Find the positive integer n that satisfies the index equation $\dfrac{3^{2n} \times 3^4}{3^5} = 3^9$.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Explain why $4^3 \times 2^5$ cannot be simplified as 8^8, and show how to rewrite 4^3 with base 2 to simplify $4^3 \times 2^5$ as a single power of 2.

Q10. Without using a calculator, determine which number is larger: 2^{30} or 3^{20}. (Hint: rewrite both using the Power of a Power Law as $(2^3)^{10}$ and $(3^2)^{10}$.)